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Determining the Free Parameters in Viable *f*(R) Models

Louis Yang, Chung-Chi Lee, Ling-Wei Luo, and Chao-Qiang Geng
Department of Physics, National Tsing Hua University, Hsinchu 300, Taiwan

Abstract

We study the observational constraints [7] on the exponential gravity model $f(R) = -\beta R_s\left(1-e^{-R/R_s}\right)$ [1]. We use the latest observational data including Supernova Cosmology Project (SCP) Union2 compilation, Two-Degree Field Galaxy Redshift Survey (2dFGRS), Sloan Digital Sky Survey Data Release 7 (SDSS DR7) and Seven-Year Wilkinson Microwave Anisotropy Probe (WMAP7) in our analysis. From these observations, we obtain a lower bound on the model parameter β at 1.27 (95% CL) but no appreciable upper bound. The constraint on the present matter density parameter is $0.245 < \Omega_m^0 < 0.311$ (95% CL). We also find out the best-fit value of model parameters on several cases.

I. Introduction

Cosmic observations from type Ia supernovae (SNe Ia), large scale structure (LSS), baryon acoustic oscillations (BAO) and cosmic microwave background (CMB) indicate that our universe is undergoing an accelerating expansion. The reason for this acceleration, the so-called dark energy problem, remains a fascinating question today.

We use the experimental constraints to determine the free parameters in viable $f(R)$ models, in our case, we studied a viable model so called the ``exponential gravity'' that proposed by E. V. Linder.

II. Exponential Gravity

The action of the exponential gravity is

$$S = \frac{1}{2\kappa^2}\int d^4x\sqrt{-g}\left[R + f(R)\right] + S_m,$$

where $\kappa \equiv 8\pi G$ and

$$f(R) = -\beta R_s\left(1-e^{-R/R_s}\right).$$

The modified Friedmann equation is

$$H^2 = \frac{\kappa^2\rho_M}{3} + \frac{1}{6}\left(f_R R - f\right) - H^2\left(f_R + f_{RR}R'\right),$$

where a subscript R denotes the derivative with respect to R, a prime represents $\mathrm{d/d\ ln}\ a$, and $\rho_M = \rho_m + \rho_r$. In flat spacetime, the Ricci scalar is

$$R = 12H^2 + 6HH'.$$

Following Hu and Sawicki's parameterization

$$y_H \equiv \frac{\rho_{DE}}{\rho_m^0} = \frac{H^2}{m^2} - a^{-3},\ y_R \equiv \frac{R}{m^2} - 3a^{-3},$$

with $m^2 \equiv \kappa^2\rho_m^0/3$, we rewrite the modified Friedmann equation and Ricci scalar into two coupled differential equations

$$y_H' = \frac{y_R}{3} - 4y_H$$

and

$$y_R' = 9a^{-3} - \frac{1}{H^2 f_{RR}}\left[y_H + f_R\left(\frac{H^2}{m^2} - \frac{R}{6m^2}\right) + \frac{f}{6m^2}\right],$$

Leading to a second order differential equation of y_H in the form

$$y_H'' + J_1 y_H' + J_2 y_H + J_3 = 0.$$

And the effective dark energy equation of state w_{DE} is given by

$$w_{DE} = -1 - \frac{y_H'}{3y_H}.$$

III. Observation Constraints

A. Type Ia Supernovae (SNe Ia)

$$\mu_{th}(z_i) \equiv 5\log_{10} D_L(z_i) + \mu_0,\ \mu_0 = 42.38 - 5\log_{10} h,\ H_0 = h\cdot 100km/s/Mpc$$

$$D_L(z) = (1+z)\int_0^z \frac{dz'}{E(z')},\ E(z) = H(z)/H_0 \Rightarrow \chi_{SN}^2 = \sum_i \frac{\left[\mu_{obs}(z_i) - \mu_{th}(z_i)\right]^2}{\sigma_i^2}.$$

B. Baryon Acoustic Oscillations (BAO)

$$D_V(z) \equiv \left[(1+z)^2 D_A(z)\frac{z}{H(z)}\right]^{1/3},\ D_A(z) = \frac{1}{1+z}\int_0^z \frac{dz'}{H(z')},$$

$$r_s(z) = \frac{1}{\sqrt{3}}\int_0^{1/(1+z)} \frac{da}{a^2 H\left(z' = \frac{1}{a} - 1\right)\sqrt{1 + \left(3\Omega_b^0/4\Omega_\gamma^0\right)a}},$$

$$z_d = \frac{1291\left(\Omega_m^0 h^2\right)^{0.251}}{1 + 0.659\left(\Omega_m^0 h^2\right)^{0.828}}\left[1 + b_1\left(\Omega_b^0 h^2\right)^{b_2}\right],$$

$$b_1 = 0.313\left(\Omega_m^0 h^2\right)^{-0.419}\left[1 + 0.607\left(\Omega_m^0 h^2\right)^{0.674}\right],\ b_2 = 0.238\left(\Omega_m^0 h^2\right)^{0.233},$$

$$\chi_{BAO}^2 = \left(x_{i,BAO}^{th} - x_{i,BAO}^{obs}\right)\left(C_{BAO}^{-1}\right)_{ij}\left(x_{j,BAO}^{th} - x_{j,BAO}^{obs}\right),\ x_{i,BAO} = \left(d_{0.2}, d_{0.35}\right).$$

C. Cosmic Microwave Background (CMB)

$$l_A(z_*) \equiv (1+z_*)\frac{\pi D_A(z_*)}{r_s(z_*)},\ R(z_*) \equiv \sqrt{\Omega_m^0} H_0 (1+z_*) D_A(z_*),$$

$$z_* = 1048\left[1 + 0.00124\left(\Omega_b^0 h^2\right)^{-0.738}\right]\left[1 + g_1\left(\Omega_b^0 h^2\right)^{g_2}\right],$$

$$g_1 = \frac{0.0783\left(\Omega_b^0 h^2\right)^{-0.238}}{1 + 39.5\left(\Omega_b^0 h^2\right)^{0.763}},\ g_2 = \frac{0.560}{1 + 21.1\left(\Omega_b^0 h^2\right)^{1.81}},$$

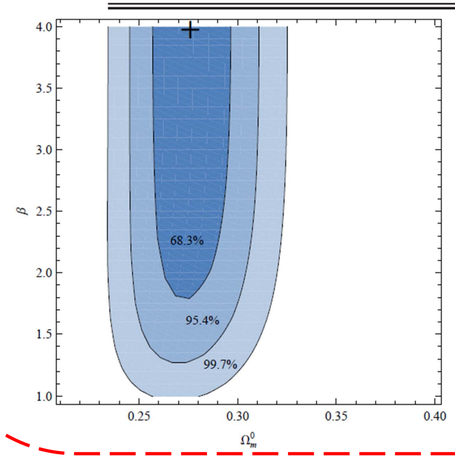
$$\chi_{CMB}^2 = \left(x_{i,CMB}^{th} - x_{i,CMB}^{obs}\right)\left(C_{CMB}^{-1}\right)_{ij}\left(x_{j,CMB}^{th} - x_{j,CMB}^{obs}\right),\ x_{i,CMB} = \left(l_A(z_*), R(z_*), z_*\right).$$

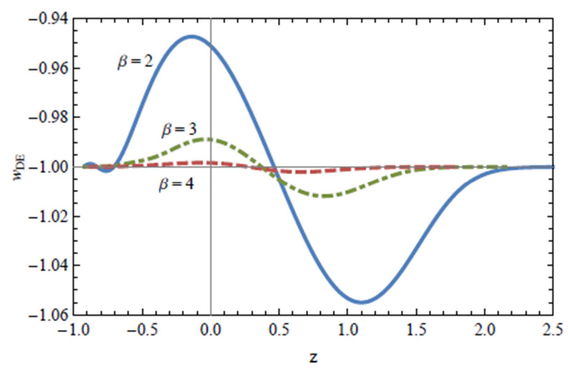
$$\Rightarrow \boxed{\chi^2 = \chi_{SN}^2 + \chi_{BAO}^2 + \chi_{CMB}^2}.$$

IV. Results

TABLE I. The best-fit values of the matter density parameter Ω_m^0 (68% CL) and χ^2 for the exponential gravity model with $\beta = 2, 3, 4$ and the ΛCDM model. Note that the error for Ω_m^0 is obtained when β is fixed.

Model		Ω_m^0	χ^2
Exponential Gravity	$\beta = 2$	$0.274^{+0.014}_{-0.013}$	546.7136
	$\beta = 3$	$0.276^{+0.014}_{-0.013}$	545.3836
	$\beta = 4$	$0.276^{+0.014}_{-0.013}$	545.1721
ΛCDM		$0.276^{+0.014}_{-0.013}$	545.1522





Reference

[1] E. V. Linder, Phys. Rev. D 80, 123528 (2009).
[2] W. Hu and I. Sawicki, Phys. Rev. D 76, 064004 (2007).
[3] R. Amanullah et al., Astrophys. J. 716, 712 (2010).
[4] W. J. Percival et al., Mon. Not. Roy. Astron. Soc. 401, 2148 (2010).
[5] E. Komatsu et al., arXiv:1001.4538 [astro-ph.CO].
[6] K. Bamba, C. Q. Geng and C. C. Lee, arXiv:1007.0482 [astro-ph.CO].
[7] Submit to Phys. Rev. D